

A Technical Introduction to Ridge regression

$$L(\beta) = \|y - X\beta\|^2 + \lambda \|\beta\|^2$$

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

Section 1

and is zero elsewhere. This demonstrates the effect of the Tikhonov parameter on the condition number of the regularized problem. For the generalized case, a similar representation can be derived using a generalized singular-value decomposition. Finally, it is related to the Wiener filter:

Section 2

$$\mathbb{E} \|Y' - X\beta\|^2 \geq \mathbb{E} \|Y' - X\hat{\beta}\|^2 + \text{tr}(\Omega)$$

$$\mathbb{E} \|Y' - X\hat{\beta}\|^2 \geq \mathbb{E} \|Y' - X\hat{\beta}\|^2 + \frac{\text{tr}(\Omega)}{\lambda}$$

Section 3

Generalized Tikhonov regularization For general multivariate normal distributions for x and the data error, one can apply a transformation of the variables to reduce to the case above. Equivalently, one can seek an x to minimize

Section 4

where $A = (A \Gamma)$ and $b = (b \ 0)$, for some suitably chosen Tikhonov matrix Γ . In many cases, this matrix is chosen as a scalar multiple of the identity matrix ($\Gamma = \alpha I$), giving preference to solutions with smaller norms; this is known as L2 regularization. In other cases, high-pass operators (e.g., a difference operator or a weighted Fourier operator) may be used to enforce smoothness if the underlying vector is believed to be mostly continuous. This regularization improves the conditioning of the problem, thus enabling a direct numerical solution. Treating it as an ordinary least squares problem with augmented matrices A and b , the solution is

Section 5

Tikhonov regularization for linear equations Suppose that for a known real matrix A and vector b , we wish to find a vector x such that

Section 6

with the "regularisation matrix" $B = (A^T A + \Gamma^T \Gamma)^{-1} A^T A$. If the parameter fit comes with a covariance matrix of the estimated parameter uncertainties V_0 , then the regularisation matrix will be

Section 7

Further reading Gruber, Marvin (1998). Improving Efficiency by Shrinkage: The James–Stein and Ridge Regression Estimators. Boca Raton: CRC Press. ISBN 0-8247-0156-9. Kress, Rainer (1998). "Tikhonov Regularization". Numerical Analysis. New York: Springer. pp. 86–90. ISBN 0-387-98408-9. Press, W. H.; Teukolsky, S. A.; Vetterling, W. T.; Flannery, B. P. (2007). "Section 19.5. Linear Regularization Methods". Numerical Recipes: The Art of Scientific Computing (3rd ed.). New York: Cambridge University Press. ISBN 978-0-521-88068-8. Saleh, A. K. Md. Ehsanes; Arashi, Mohammad; Kibria, B. M. Golam (2019). Theory of Ridge Regression Estimation with Applications. New York: John Wiley & Sons. ISBN 978-1-118-64461-4. Taddy, Matt (2019). "Regularization". Business Data Science: Combining Machine Learning and Economics to Optimize, Automate, and Accelerate Business Decisions. New York: McGraw-Hill. pp. 69–104. ISBN 978-1-260-45277-8.

Section 8

Relation to probabilistic formulation The probabilistic formulation of an inverse problem introduces (when all uncertainties are Gaussian) a covariance matrix C_M representing the a priori uncertainties on the model parameters, and a covariance matrix C_D representing the uncertainties on the observed parameters. In the special case when these two matrices are diagonal and isotropic, $C_M = \sigma_M^2 I$ and $C_D = \sigma_D^2 I$, and, in this case, the equations of inverse theory reduce to the equations above, with $\alpha = \sigma_D^2 / \sigma_M^2$.

Section 9

the problem of a near-singular moment matrix $X^T X$ is alleviated by adding positive elements to the diagonals, thereby decreasing its condition number. Compared to the ordinary least squares estimator, the simple ridge estimator

has an extra term $\lambda \mathbf{I}$ in the denominator:

Section 10

which shows that λ is nothing but the Lagrange multiplier of the constraint. In fact, there is a one-to-one relationship between c and λ and since, in practice, we do not know c , we define λ heuristically or find it via additional data-fitting strategies, see Determination of the Tikhonov parameter below. Note that as $\lambda \downarrow 0$, the constraint eventually becomes non-binding, and the ridge estimator converges to the minimum-norm ordinary least squares estimator, here denoted as $\hat{\beta} = \beta^0$:

Section 11

where the Wiener weights are $f_i = \frac{\sigma_i^2}{\sigma_i^2 + \alpha^2}$ and q is the rank of A .

Section 12

where $\mathbf{x}$ and $\mathbf{b}$ may be of different sizes and A may even be non-square. The standard approach is ordinary least squares linear regression. However, if no $\mathbf{x}$ satisfies the equation or more than one $\mathbf{x}$ does—that is, the solution is not unique—the problem is said to be ill posed. In such cases, ordinary least squares estimation leads to an overdetermined, or more often an underdetermined system of equations. Most real-world phenomena have the effect of low-pass filters in the forward direction where A maps $\mathbf{x}$ to $\mathbf{b}$. Therefore, in solving the inverse-problem, the inverse mapping operates as a high-pass filter that has the undesirable tendency of amplifying noise (eigenvalues / singular values are largest in the reverse mapping where they were smallest in the forward mapping). In addition, ordinary least squares implicitly nullifies every element of the reconstructed version of $\mathbf{x}$ that is in the null-space of A , rather than allowing for a model to be used as a prior for $\mathbf{x}$. Ordinary least squares seeks to minimize the sum of squared residuals, which can be compactly written as